

Lab: Experimental Design and Statistical Testing

Name: Martin Lenga

Introduction

As a data analyst at Spotify, the goal of this experiment is to determine whether a new personalized playlist algorithm improves weekly listening time compared to the current recommendation system. This lab applies concepts of hypothesis testing, experimental design, statistical significance, business decision-making, and interpretation of results.

Part 1: Research Question Definition and Hypothesis Formation

Hypotheses

i) Null Hypothesis (H_0)

The new personalized playlist algorithm does not increase weekly listening time compared to the current algorithm.

ii) Alternative Hypothesis (H_1)

The new personalized playlist algorithm increases weekly listening time compared to the current algorithm.

Reflection Questions

1. What makes this initial request problematic for hypothesis testing?

The request is vague because terms such as “listen more” and “better at finding music” are not clearly measurable. The request does not specify:

- The exact metric to measure
- The target users
- The testing duration
- The amount of expected improvement
- The business success criteria

Without these details, it is difficult to design a valid statistical test.

2. How would you refine this into a specific, measurable research question?

A refined research question would be:

“Does the new personalized playlist algorithm increase average weekly listening time per user compared to the current algorithm over a two-week testing period?”

This question is measurable, specific and appropriate for hypothesis testing.

3. What additional information would you need from the product team?

Additional information needed includes:

- Definition of user engagement
- Expected minimum improvement
- Current average listening time
- Target user groups
- Rollout timeline
- Business cost of implementation
- Secondary success metrics such as retention or user satisfaction

4. Would a one-sided test or two-sided test be best in this scenario? Why?

A one-sided test would be appropriate because the product team specifically expects the new algorithm to increase listening time. The focus is only on improvement rather than any change in either direction.

5. What are the potential risks of choosing a one-sided test?

Potential risks include:

- Missing negative effects if listening time decreases
- Introducing bias toward positive findings
- Reducing credibility of results
- Ignoring unintended consequences

A one-sided test should only be used when decreases are not relevant to the business decision.

Part 2: Experimental Design Framework

Reflection Questions

1. What factors might affect listening time that need to be controlled for?

Several factors may influence listening time and should be controlled:

- Subscription type (free vs premium)
- Geographic location
- Device type
- User demographics
- Internet accessibility
- Existing listening habits
- Music preferences
- Seasonal behavior

Controlling these variables helps ensure the algorithm is the main factor affecting outcomes.

2. How would you ensure representative sampling across user segments?

Representative sampling can be achieved through stratified random sampling.

Users should be proportionally selected based on:

- Subscription type
- Geographic region
- Device usage

For example:

- 40% premium users
- 60% free-tier users
- Geographic proportions matching the total user base

Random assignment within each subgroup ensures balanced representation.

3. What potential biases could emerge from improper sampling?

Potential biases include:

- Overrepresentation of premium users
- Geographic bias
- Device usage bias
- Selection bias

- Survivorship bias

Improper sampling may produce misleading conclusions and reduce experiment validity.

Part 3: Setting Decision Criteria

Reflection Questions

1. How would the implementation cost affect your choice of significance level (α), minimum effect size and statistical power?

i) Significance Level (α)

Because implementation costs are high, a stricter significance level such as $\alpha = 0.01$ may be used to reduce the chance of false positives.

ii) Minimum Effect Size

The business should define a meaningful improvement threshold, such as:

- At least 5–10 additional listening minutes per week

Very small improvements may not justify engineering and operational costs.

iii) Statistical Power

A high statistical power such as 80% or 90% is important to reduce the risk of failing to detect a true improvement.

Higher power generally requires larger sample sizes.

2. What specific business metrics would you need to define “significant improvement”?

Important business metrics include:

- Weekly listening time
- Daily active users
- Retention rate
- Subscription conversions
- Session duration
- Playlist completion rate
- Revenue impact

A result should be both statistically and practically significant.

3. What secondary metrics could be monitored for potential negative impacts?

Secondary metrics may include:

- User churn rate
- App uninstall rate
- Song skip rate
- User complaints
- Battery usage
- Data consumption
- User satisfaction ratings

Monitoring these metrics helps identify unintended side effects.

Part 4: Data Collection

Experiment Results

i) **Control Group**

- Sample size: 500,000 users
- Mean listening time: 118.5 minutes
- Standard deviation: 44.2 minutes
- 95% Confidence Interval: [117.8, 119.2]

ii) **Treatment Group**

- Sample size: 500,000 users
- Mean listening time: 122.3 minutes
- Standard deviation: 43.8 minutes
- 95% Confidence Interval: [121.6, 123.0]

Reflection Questions

1. What do the experiment results suggest?

The treatment group achieved a higher average listening time than the control group.

Difference in means:

$122.3 - 118.5 = 3.8$ minutes

This suggests that the new algorithm may improve user engagement.

2. What does the confidence interval indicate?

The confidence intervals show the likely range for the true population mean.

Since the intervals show minimal overlap, this suggests the observed increase is likely statistically meaningful rather than due to random chance.

3. Why is randomization important?

Randomization helps:

- Reduce bias
- Balance confounding variables
- Ensure fair comparison between groups
- Improve validity of conclusions

Without randomization, observed differences may not be caused by the algorithm itself.

Part 5: Analyze and Interpret

Scenario 1

Given Results

- p-value = 0.047
- Average listening time increase = 8 minutes per week
- Sample size = 500,000 users per group
- Test duration = 2 weeks

Reflection Questions

1. How would you explain this p-value to non-technical stakeholders?

A p-value of 0.047 means that if the new algorithm truly had no effect, there would only be a 4.7% chance of observing results this extreme due to random chance.

Since the p-value is below 0.05, the result is considered statistically significant.

However, stakeholders should understand that:

- A p-value does not prove the algorithm works with certainty

- It does not measure the size of the improvement
- It does not guarantee future results

Other considerations include:

- Business value
- Implementation cost
- Long-term user engagement
- User satisfaction

2. If 5 million users per group returned $p = 0.001$ but only a 2-minute increase, how would this change your interpretation?

With very large sample sizes, even very small differences can become statistically significant.

Although $p = 0.001$ suggests strong statistical evidence, a 2-minute increase may not be practically meaningful enough to justify implementation costs.

This demonstrates the difference between:

- Statistical significance
- Practical significance

Business impact should always be considered alongside statistical results.

Scenario 2

Given Results

- $p\text{-value} = 0.08$
- Effect size = 12 minutes increased listening time
- High variance in data
- Strong positive trends in user satisfaction

Reflection Questions

1. How would you approach making a recommendation in this case?

Although the result is not statistically significant at the traditional 0.05 level, the large effect size and positive user satisfaction trends suggest the algorithm may still provide business value.

I would recommend:

- Running the experiment longer
- Collecting additional data
- Investigating causes of high variance
- Performing subgroup analysis

A decision should not rely solely on the p-value.

2. How would you handle stakeholder pressure to make a definitive “yes/no” decision based solely on statistical significance?

I would explain that statistical significance is only one part of decision-making.

Important business decisions should also consider:

- Effect size
- User satisfaction
- Revenue impact
- Operational costs
- Long-term retention

A strict yes/no interpretation may ignore important context.

3. What additional data or analyses might help provide more context for decision-making?

Additional analyses may include:

- Longer testing duration
- Segment-specific analysis
- Retention analysis
- Revenue impact analysis
- Confidence interval analysis
- Variance investigation
- User satisfaction surveys
- Regional A/B testing

These analyses would provide deeper insight into the algorithm’s overall business impact.

Conclusion

The experiment demonstrates how hypothesis testing and experimental design can support business decision-making at Spotify. While statistical significance is important, practical significance, implementation cost, user satisfaction, and long-term business value must also be considered before rolling out a new recommendation algorithm globally.